Example 1.17 If $f : \mathbb{R} - \{-1, 1\} \to \mathbb{R}$ is defined by $f(x) = \frac{x}{x^2-1}$, verify whether f is one-to-one or not.

Solution:
We start with the assumption $f(x) = f(y)$. Then,

$$\begin{array}{rrcl} & \frac{x}{x^2-1} & = & \frac{y}{y^2-1} \\ \Rightarrow & x(y^2-1) & = & y(x^2-1) \\ \Rightarrow & xy^2 - x - yx^2 + y & = & 0 \\ \Rightarrow & (y-x)(xy+1) & = & 0 \end{array}$$

This implies that $x = y$ or $xy = -1$. So if we select two numbers x and y so that $xy = -1$, then $f(x) = f(y)$. $(2, -\frac{1}{2}), (7, -\frac{1}{7}), (-2, \frac{1}{2})$ are some among the infinitely many possible pairs. Thus $f(2) = f(\frac{-1}{2}) = \frac{2}{3}$. That is, $f(x) = f(y)$ does not imply $x = y$. Hence it is not one-to-one.

Example 1.18 If $f : \mathbb{R} \to \mathbb{R}$ is defined as $f(x) = 2x^2 - 1$, find the pre-images of $17, 4$ and -2.

Solution:
To find the pre-image of 17, we solve the equation $2x^2 - 1 = 17$. The two solutions of this equation, 3 and -3 are the pre-images of 17 under f. The equation $2x^2 - 1 = 4$ yields $\sqrt{\frac{5}{2}}$ and $-\sqrt{\frac{5}{2}}$ as the two pre-images of 4. To find the pre-image of -2, we solve the equation $2x^2 - 1 = -2$. This shows that $x^2 = -\frac{1}{2}$ which has no solution in $\mathbb{R}$ because square of a number cannot be negative and hence -2 has no pre-image under f.

Example 1.19 If $f : [-2, 2] \to B$ is given by $f(x) = 2x^3$, then find B so that f is onto.

Solution:
The minimum value is $f(-2)$ and its maximum value is $f(2)$ which are equal to -16 and 16 respectively. So B is $[-16, 16]$.



As $f(x) = 2x^3$ is an increasing function on $[-2, 2]$, the minimum value is attained at the left end and the maximum value is attained at the right end. (For more about increasing / decreasing functions one may refer later chapters.)

Example 1.20 Check whether the function $f(x) = x|x|$ defined on $[-2, 2]$ is one-to-one or not. If it is one-to-one, find a suitable co-domain so that the function becomes a bijection.

Solution:
Let $x, y \in [-2, 2]$ such that $f(x) = f(y)$. If $y = 0$, then $x = 0$. So let $y \neq 0$ and hence $x \neq 0$. Now $x|x| = y|y|$ since $f(x) = f(y)$. This implies that $\frac{x}{y} = \frac{|y|}{|x|}$. Since $\frac{|y|}{|x|} > 0, \frac{x}{y} > 0$; thus x and y are either both positive or both negative and hence $x^2 = y^2$.

So if $f(x) = f(y)$, we must have $x^2 = y^2$. Also x and y are either both negative or both positive. This is possible only if $x = y$. Thus f is one-to-one. When $x < 0$, $f(x) = -x^2$ and when $x \geq 0, f(x) = x^2$. So the range is $[-4, 4]$. So f becomes a bijection from $[-2, 2]$ to $[-4, 4]$.